

GPUREC: A GPU-Accelerated, Differentiable Framework for Gene–Species Tree Reconciliation under the DTL Model with Certified Optima, Flexible Priors, and Transfer Heterogeneity

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Abstract

Reconciling gene trees with a species tree under a model of gene *duplication, transfer, and loss* (DTL) is central to comparative genomics. Existing reconciliation software — including the state-of-the-art ALERAX — treats the amalgamated likelihood as a fixed, hand-differentiated black box, optimized by first-order methods on the CPU. We re-implement the UNDATEDDTL reconciliation likelihood as a fully *differentiable* GPU program, exposed through a PYTORCH interface. Our central claim is methodological: *differentiable reconciliation is an enabling computational paradigm* — exact gradients, Hessian–vector products, the information matrix, uncertainty quantification, custom priors, and model extensions all follow from it, essentially for free. Concretely, this single capability yields (i) order-of-magnitude faster maximum-likelihood estimation of per-family (“genewise”) DTL rates; (ii) *second-order* inference — certified *local* optimality via the Hessian spectrum, and the observed-information matrix for parameter uncertainty and non-identifiability analysis; (iii) *arbitrary user-defined penalties*, demonstrated by cross-validating a tree-smoothing prior for species-wise rates on HOGENOM, a computation that would be infeasible on the CPU; and (iv) *per-recipient transfer weights* that infer which species disproportionately receive horizontal transfers, relaxing the uniform-recipient assumption and addressing a feature listed as future work for ALERAX. We demonstrate on the HOGENOM-CORE and archaeal datasets, and release GPUREC as an open, extensible engine. **[TODO: 1–2 sentences of headline quantitative results once experiments land.]**

1 Introduction

Genomes record evolution at multiple scales: substitutions accumulate along gene trees, while gene duplication, horizontal transfer, and loss reshape gene family histories relative to the species tree (Szöllősi et al., 2015). Accounting for this discord is the goal of probabilistic *reconciliation* under the DTL model. Because individual gene trees are themselves uncertain, modern methods integrate over a *distribution* of gene trees summarised by conditional clade probabilities (CCPs), as in Amalgamated Likelihood Estimation (Szöllősi et al., 2013; Höhna and Drummond, 2012; Larget, 2013). GENERAX (Morel et al., 2020), SPECIESRAX (Morel et al., 2022), and most recently ALERAX (Morel et al., 2024) build on this framework to co-estimate reconciled gene trees and a shared species tree under the UNDATEDDTL model.

ALERAX is the current state of the art for amalgamated DTL reconciliation. It estimates the model’s rate parameters — per branch e , a duplication, transfer, and loss probability $(\delta_e, \tau_e, \lambda_e)$ —

by *gradient descent* on the amalgamated likelihood, and supports three parameter-sharing regimes: *global* (all families and branches share (δ, τ, λ) , 3 parameters), *per-family* ($3K$ parameters for K families), and *per-species* ($3n$ parameters for n user-defined species groupings) (Morel et al., 2024). It runs an order of magnitude faster than ALE and scales to genome-wide datasets with hundreds of taxa.

Why ALERAX is our primary baseline. The amalgamated-DTL lineage comprises ALE (Szöllősi et al., 2013), which scores a fixed species tree against gene-tree distributions; GENERAX (Morel et al., 2020), which infers reconciled gene trees from *single* maximum-likelihood gene trees (no gene-tree-distribution input); SPECIESRAX (Morel et al., 2022), which targets species-tree inference; and ALERAX (Morel et al., 2024), which unifies these by estimating DTL rates and reconciliations from gene-tree *distributions* with the same per-family/per-species rate-sharing modes we adopt, an order of magnitude faster than ALE. Because ALERAX is the most recent, fastest, and most directly comparable tool for the *rate-estimation* task we accelerate — same likelihood, same parameter-sharing regimes, same CCP input — it is the natural head-to-head baseline; we benchmark against it throughout and reference the others where the comparison is informative.

Despite these strengths, the rate-estimation engine of existing tools has structural limitations that motivate this work:

1. **First-order, CPU-bound optimization.** Rates are fit with gradient descent on the CPU. Without curvature information there is no way to *certify* that an estimate is a (local) optimum, to quantify the *uncertainty* of the inferred rates, or to detect when parameters are *non-identifiable* (statistically confounded).
2. **A fixed model.** The likelihood, recipient distribution for transfers, and the (absence of a) prior are baked in. Incorporating biologically motivated structure — e.g. smoothing rates across the species tree, or letting some lineages be preferential transfer recipients — is not readily available to the user. Morel et al. (2024) explicitly list “DTL rate heterogeneity over species” among directions for future work.
3. **Scale limits on model selection.** Choosing a regularization strength by cross-validation, or running many restarts, multiplies the cost of an already expensive fit, which is impractical on the CPU.

Contribution. We re-implement the UNDATEDDTL amalgamated reconciliation likelihood as a differentiable GPU program with a PYTORCH front-end, GPUREC. The implementation provides exact gradients (via implicit differentiation of the model’s fixed-point solves) and exact Hessian–vector products (via forward-over-reverse differentiation), entirely matrix-free. From this single capability we obtain, without bespoke derivations:

1. **Speed.** GPU evaluation makes per-family rate estimation embarrassingly parallel across families, giving an order-of-magnitude speedup over CPU optimization (Sec. 3.1).
2. **Certified optima and uncertainty.** Hessian–vector products feed a Lanczos solver that certifies a positive-definite local minimum, and yield the Fisher (observed) information matrix for asymptotic confidence intervals and identifiability analysis (Secs. 3.2–3.3).
3. **Flexible priors / penalties.** Any differentiable penalty $R(\theta)$ can be added to the loss; we cross-validate a tree-smoothing prior for species-wise rates on HOGENOM (Sec. 3.5).
4. **Transfer heterogeneity.** Learned per-recipient weights identify species that receive more transfers than others (Sec. 3.4).

Scope. We focus on the *parameter-estimation / reconciliation-likelihood* component: given a fixed species tree and per-family CCP distributions, infer DTL rates (and reconciliations). Species-tree search, as performed by SPECIESRAX/ALERAX, is orthogonal and could be driven by the same differentiable engine; we return to this in Sec. 4.

2 Model and Methods

2.1 The amalgamated UNDATEDDTL likelihood

Let S be a rooted species tree and, for each gene family i , let A_i be a sample of gene trees summarised by its CCP distribution. Under amalgamation, the family likelihood marginalises over all gene trees compatible with the CCPs (Szöllősi et al., 2013):

$$P(A_i | S) = \sum_G P(A_i | G) P(G | S), \quad (1)$$

and the dataset likelihood factorises over families, $\mathcal{L}(S | A) = \prod_i P(A_i | S)$. Under UNDATEDDTL, a gene lineage on species-tree branch e undergoes duplication, transfer, and loss with per-branch probabilities $(\delta_e, \tau_e, \lambda_e)$; survival probabilities are obtained from a fixed-point (“extinction”) equation, and $P(A_i | S)$ from a dynamic-programming recursion over CCP clades (Szöllősi et al., 2013; Morel et al., 2020). We collect the free log-rate parameters into a vector θ and write the negative log-likelihood $\mathcal{L}(\theta) = -\sum_i \log P(A_i | S; \theta)$.

2.2 Parameter-sharing modes

GPUREC supports the same sharing regimes as ALERAX: *global* ($\theta \in \mathbb{R}^3$), *genewise* (per-family, $\theta \in \mathbb{R}^{3K}$), and *specieswise* (per-species or per-grouping, $\theta \in \mathbb{R}^{3n}$; $p = 3n$). The genewise mode reproduces the per-family optimization of ALERAX and is the natural target for the speed comparison; the specieswise mode is the setting in which priors and second-order analysis are most informative.

2.3 A differentiable GPU implementation

The likelihood is implemented as a PYTORCH tensor program. The two inner solves — the extinction fixed point and the per-clade survival/transfer recursion — are iterative, and we differentiate them by *implicit differentiation* rather than by differentiating a truncated unrolled loop. *Hessian-vector products* $\mathbf{H}(\theta)v$ are obtained matrix-free by forward-over-reverse differentiation, so second-order information costs a small constant number of solves per product and never forms the $p \times p$ Hessian. Throughput is sustained by batching families across the GPU and, optionally, by adapting per-batch solver fidelity (the number of fixed-point / Neumann iterations) to each batch’s convergence, so that “easy” families are not over-solved.

Why implicit differentiation is correct. Let $x^*(\theta)$ be the converged solution of the model’s fixed-point map,

$$x^*(\theta) = F(x^*(\theta), \theta), \quad (2)$$

(the extinction probabilities, and similarly the clade recursion). By the implicit function theorem, differentiating Eq. (2) gives

$$\frac{dx^*}{d\theta} = (I - \partial_x F)^{-1} \partial_\theta F, \quad (3)$$

valid wherever $(I - \partial_x F)$ is invertible (guaranteed when the fixed-point iteration is contractive, i.e. the spectral radius of $\partial_x F$ is below one — the same condition under which the forward solve converges). The downstream loss gradient is then assembled by the chain rule. Crucially, the inverse in Eq. (3) is *never formed*: the adjoint/cotangent system $(I - \partial_x F)^\top u = b$ is solved *iteratively* by a Neumann series (equivalently, the same contraction run in reverse), so the gradient is exact up to the adjoint solver’s tolerance and costs $O(1)$ extra solves. Differentiating the unrolled forward iterations would instead make the gradient’s accuracy hostage to the (truncated) iteration count and inflate memory with the full iteration trace; implicit differentiation decouples gradient accuracy from forward-iteration depth. **[TODO: cite the implicit function theorem / adjoint method; one sentence on the Triton kernels / Rust preprocessing if desired.]**

2.4 Per-recipient transfer weights

In the standard model a transferred copy is received by a branch drawn from a fixed, effectively uniform recipient distribution. We relax this with *species-specific* recipient weights. Let $\alpha \in \mathbb{R}^S$ be a vector of per-species logits and define the recipient distribution by a softmax,

$$w_s = P(\text{recipient} = s) = \frac{\exp(\alpha_s)}{\sum_r \exp(\alpha_r)}, \quad \sum_s w_s = 1. \quad (4)$$

The simplex constraint $\sum_s w_s = 1$ removes the scale ambiguity (only relative weights are meaningful) and makes w *identifiable separately from the transfer rate* τ : τ sets the *total* probability that a lineage is transferred, while w sets only the *normalized distribution* over recipients. Large w_s flags species that act as transfer *sinks*. Because α is weakly constrained when transfers into a clade are rare, we regularize it with a smoothness/entropy prior (e.g. the same tree-Laplacian of Sec. 2.5 applied to α , or a symmetric Dirichlet on w), selected like any other penalty (Sec. 2.5). This is a concrete form of the across-species heterogeneity that Morel et al. (2024) list as future work.

Because α enters the likelihood only through the softmax of Eq. (4), the loss is exactly invariant under $\alpha \mapsto \alpha + c\mathbf{1}$ (a shift of all logits); we fix this gauge by optimizing in the mean-zero subspace. The receiver-weight analogue of the rate box of Sec. 3.2 is a floor on the recipient *probability* itself — a bound on the logits would be vacuous, since the same softmax shift that fixes the gauge can drive any $w_s \rightarrow 0$ regardless of where the logits sit. We therefore reparametrize $w = c\mathbf{1} + (1 - Sc)\text{softmax}(\beta)$ with $c = 1/(100S)$, which guarantees $w_s \geq 1/(100S)$ for every species (no recipient weight collapses) while leaving dominant transfer sinks ($w_s \rightarrow 1$) unconstrained, and optimize the unconstrained β . The exact gradient $\partial \mathcal{L} / \partial \alpha$ is produced by the *same* analytic backward that yields the rate gradient (the recipient distribution is just another differentiable input to the likelihood), and the matrix-free HVP extends to the full (θ, α) block. Gradient, (θ, α) HVP, and the gauge-projected Fisher information are validated against **fp64** finite differences (per-block relative error $\sim 10^{-6}$), so α is estimated jointly with the rates and its identifiability and standard errors come from the same matrix-free machinery, at no extra derivation cost.

2.5 Penalized likelihood and priors

Because the objective is an ordinary differentiable function in PYTORCH, the user may add *any* differentiable penalty $R(\theta)$ and optimize

$$F_\kappa(\theta) = \mathcal{L}(\theta) + \kappa R(\theta), \quad (5)$$

recovering MAP estimation when κR is the negative log-prior. We highlight two:

- **Ridge / Gaussian prior:** $R(\theta) = \frac{1}{2}\|\theta - \theta_0\|^2$.
- **Tree-smoothing (roughness) prior** in the spirit of Sanderson (2002): penalise how much species-wise rates change between adjacent species-tree branches,

$$R(\theta) = \frac{1}{2} \sum_{(c,p) \in E(S)} \|\theta_c - \theta_p\|^2, \quad (6)$$

the quadratic form of a tree graph-Laplacian. This shares statistical strength across the species tree and regularises otherwise weakly-identified per-species rates.

The penalty strength κ is a hyperparameter selected by cross-validation (Sec. 3.5).

2.6 Second-order inference

At a candidate estimate $\hat{\theta}$ with gradient $\mathbf{g} = \nabla F(\hat{\theta})$ and Hessian $\mathbf{H} = \nabla^2 F(\hat{\theta})$ (both available matrix-free):

- **Certification.** $\hat{\theta}$ is a local minimum if $\|\mathbf{g}\| \approx 0$ and $\mathbf{H} \succ 0$. We estimate the smallest eigenvalue $\lambda_{\min}(\mathbf{H})$ by Lanczos iteration on the HVP operator; a negative $\lambda_{\min}$ exhibits a descent (saddle) direction, and a positive $\lambda_{\min}$ certifies a strict local minimum. When $\lambda_{\min}$ is near zero we additionally use a true-loss line search along the lowest-curvature directions as an operational convergence test.
- **Information and uncertainty.** We use the *observed information* matrix — the Hessian of the negative log-likelihood at the optimum, $\mathbf{I}_{\text{obs}} = \mathbf{H}(\hat{\theta})$ — as the standard large-sample approximation to the Fisher information. (The observed information and the expected Fisher information $\mathbf{I}_F = \mathbb{E}[-\nabla^2 \log L]$ coincide asymptotically under the usual regularity conditions but are not literally equal; we report the observed information, which is what the matrix-free Hessian gives directly.) Its inverse $\mathbf{I}_{\text{obs}}^{-1}$ is the asymptotic covariance of the rate estimates; diagonal entries give standard errors / confidence intervals, and the eigenvectors of near-zero eigenvalues expose *confounded* parameter combinations (non-identifiability).
- **Optimization.** The same HVPs drive Newton / trust-region steps and saddle-escape for the hardest (near-singular) fits.

2.7 Optimization pipeline

A typical fit runs first-order optimization (Adam) to enter the basin, then a quasi-Newton phase (L-BFGS), and — where certification is desired — a matrix-free Newton / saddle-escape phase using the HVPs of Sec. 2.6. Convergence is judged on the projected gradient and, near singular optima, on the operational line-search criterion. **[TODO: final wording once the specieswise endgame experiment is settled.]**

3 Results

Datasets. We use the empirical datasets of Morel et al. (2024): the HOGENOM-CORE set of 666 representative genomes (Penel et al., 2009), from which we draw 1,055 gene families for the cross-validation study (Sec. 3.5), and the archaeal dataset of 5,446 gene families (≥ 4 sequences each) over $S=119$ species spanning DPANN, Euryarchaeota, TACK, and Asgardarchaeota (Williams et al., 2017; Davín et al., 2018), used for the certification and identifiability analyses ($p=3S=357$ species-wise rates).

Table 1: GPUREC genewise rate fitting on HOGENOM, restricted to families covering ≥ 4 species (the set ALERAX retains; #families is after the filter, from nominal draws of 512/1,055/all). Single RTX 4090, library recipe `gpurec.fit_genewise` ($|Pg| < 10^{-3}$, $\rho \in [10^{-6}, 2]$). Wall-clock is the optimize time; the converged count and total NLL (bits) come from an optional cold PD certificate ($\pi=64$ /Neumann=64). The ALERAX v1.4.0 row (24 CPU cores) is on the *identical* 10,869-family set, run in chunks under an 80 GB memory cap (exact, since per-family fits are independent); its NLL is converted from nats ($\div \ln 2$). GPUREC is $\sim 23\times$ faster at a marginally lower NLL.

Dataset	#families	wall-clock	converged	total NLL (bits)
HOGENOM-512	506	119 s	500/506	324,262
HOGENOM-1055	1,042	219 s	1,037/1,042	577,593
HOGENOM-full	10,869	758 s	10,853/10,869	1,906,464
ALERAX (24 CPU cores)	10,869	17,682 s	—	1,913,087

3.1 Genewise rate estimation is an order of magnitude faster

[EXPERIMENT — Speed vs. AleRax — GPUREC genewise is $\sim 23\times$ faster]

On the identical 10,869-family ≥ 4 -species set (same species tree, same gene-tree distributions, same per-family UNDATEDDTL model, same ≥ 4 -species filter), ALERAX v1.4.0 on 24 CPU cores takes 17,682 s (≈ 4.9 h) versus GPUREC’s 758 s on one RTX 4090 — a $\sim 23\times$ speed-up (Table 1, Fig. 1). GPUREC also reaches a marginally better optimum: NLL 1,906,464 vs 1,913,087 bits ($-6,623$ bits, i.e. higher likelihood) — the same model fit to a higher-likelihood point. ALERAX was run in chunks under an 80 GB memory cap (it otherwise exhausts > 300 GB loading all evaluators at once); chunking is exact because the per-family fits are independent.

GPUREC genewise throughput. Fitting independent per-family DTL rates to a converged, box-bounded optimum ($|Pg| < 10^{-3}$, $\rho \in [10^{-6}, 2]$) on HOGENOM runs entirely on a single RTX 4090: the 10,869 families covering ≥ 4 species — the set ALERAX’s UNDATEDDTL retains, making the head-to-head apples-to-apples — in ≈ 13 minutes, with 99.85% certified converged (interior positive-definite or KKT bound-active). Per-family throughput *rises* with scale — $4.3 \rightarrow 14.3$ families/s from 506 to 10,869 — as batching saturates the GPU (Table 1, Fig. 1). This recipe (Adam warm-up $\rightarrow$ box-constrained trust-region Newton on the per-family 3×3 forward-difference Hessian, with convergence-based rebatching and π -tier escalation) is the GPUREC library default, `gpurec.fit_genewise`. The adjoint warm-start that makes the cheap $\pi=16$ tier accurate carries a resident cache scaling as $\text{clades} \times \text{species} \times \text{dtype}$ — the dominant allocation, ≈ 19 GiB on the full set; a memory-aware gate keyed on that product runs the large early batches cold and re-enables the warm-start as the active set shrinks, keeping the full dataset within the 24 GB device with no manual tuning.

3.2 Certified optimality, and why the unbounded MLE is ill-posed

We are careful with wording: a positive smallest eigenvalue $\lambda_{\min}(\mathbf{H}) > 0$ certifies a *strict local* minimum only. It does *not* establish global optimality, biological correctness, or the absence of other minima. With that scope, the test is a genuine guarantee that CPU first-order tools cannot provide — and, just as usefully, it *exposes* when a reported estimate is not a minimum at all.

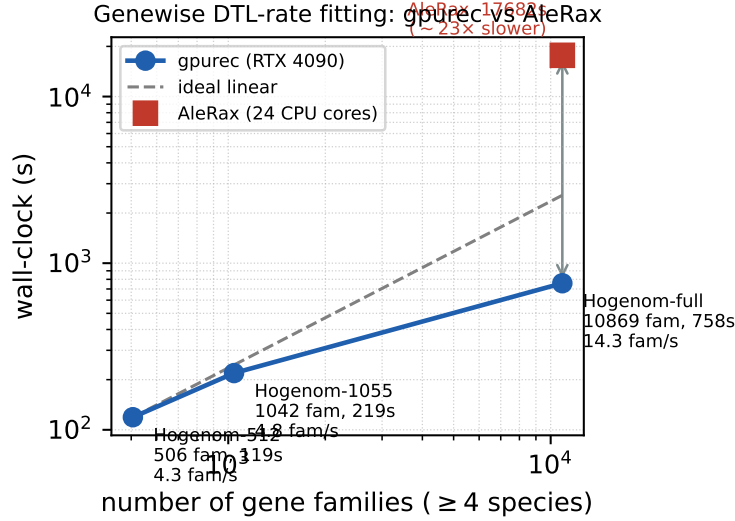


Figure 1: Genewise DTL-rate fitting wall-clock vs. number of gene families on HOGENOM (single RTX 4090; log-log, dashed = ideal linear scaling). Throughput improves with scale as batching saturates the GPU. The ALERAX CPU baseline ([**TODO:**]) will be overlaid for the head-to-head.

The unbounded species-wise MLE does not converge to a minimum. Maximum-likelihood fitting of per-species rates with only a light smoothing penalty ($\kappa=0.03$) on the full archaeal data is ill-posed. For many species the likelihood is flat in a *turnover* direction (Sec. 3.3): about 38% of the $p=357$ rates run off to the edge of any plausible range ($\hat{\rho} \gtrsim 33$, i.e. $\log_2 \hat{\rho} > 5$), because once a rate saturates the likelihood becomes insensitive to it. This boundary runaway *is* the near-singular Hessian direction — the conditioning is $\kappa(\mathbf{H}) \approx 3.8 \times 10^4$ — so first-order and quasi-Newton optimizers crawl and stop at a point that *looks* converged by loss but is not stationary: an **fp64** gradient check at the best coarse-to-fine estimate (NLL 357,039) returns $\|\mathbf{g}\| = 5.25$, far from zero. The loss alone cannot reveal this; the gradient norm and Hessian spectrum do. This is the generic hazard of unbounded rate parameters in DTL reconciliation, and it is precisely what a curvature-aware, certifying optimizer is for.

Bounding the rates restores a certifiable minimum. The remedy is to declare a box on the rates, $\rho \in [0.05, 2.0]$ (equivalently $\theta \in [\log_2 0.05, \log_2 2]$), which pins the runaway directions as active constraints and removes the near-singular subspace. A *free-subspace* projected trust-region Newton-CG — the box-active coordinates are masked out of the CG operator and the gradient, so steps never push them into the walls — run in **fp64** with an integrated negative-curvature escape, then converges to a certified bound-constrained KKT point. On all 5,446 archaeal families: NLL 359,591.9, projected-gradient norm $\|\mathbf{Pg}\| = 3.0 \times 10^{-3}$, and a *positive-definite reduced Hessian* ($\lambda_{\min} = +0.28$ on the 187 free coordinates, Lanczos Ritz residual 2×10^{-3}), with all 170/357 active-set signs KKT-correct (no escape steps needed). The bounded optimum costs $\approx 2,550$ NLL relative to the non-stationary 357,039: exactly the overfit the boundary runaway was buying. Crucially, the active set is not arbitrary — it *is* the list of weakly-identified rates (Sec. 3.3), so the box is a declared, interpretable regularizer for the directions the data cannot pin, not a numerical convenience. A penalty (Sec. 3.5) is the smooth alternative: at the cross-validated $\kappa^* \approx 1$ the smoothing itself removes the soft directions and the *unconstrained* minimum is already PD ($\lambda_{\min} = +0.22$).

3.3 Fisher information: uncertainty and identifiability

This is, we believe, the most novel result of the paper: reconciliation tools report point estimates, almost never uncertainty, confidence intervals, or identifiability structure. From the same matrix-free second-order machinery we form the exact observed-information matrix $\mathbf{I}_{\text{obs}}$ at a species-wise optimum ($p = 3S = 357$, so a dense `eigh` is cheap), read off asymptotic standard errors from $\mathbf{I}_F^{-1}$, and diagnose the confounded directions from its spectrum — an analysis impossible with first-order software, and one that directly motivates the regularization of Sec. 3.5.

What the data cannot separate: duplication vs. loss. On the archaeal species-wise fit the observed information $\mathbf{I}_{\text{obs}}$ is itself *indefinite* — 4 negative eigenvalues and 90 of 357 below 0.1 (Fig. 2, top-left) — so a quarter of the directions are left unconstrained by the data and only the prior makes the penalized Hessian positive definite. The poorly-identified directions are interpretable. Per species the confounded pair is *duplication and loss* (normalized curvature coupling 0.93, 100% of it within-species): the turnover combination $\delta + \lambda$ is soft (duplication and loss are anti-correlated in the likelihood — a copy gained then lost leaves a similar pattern to no event), while the net direction $\delta - \lambda$ is stiff and well-determined. Transfer τ is statistically *decoupled* from both (off-diagonal curvature ≈ 0) and so separately identifiable, but it is the most weakly curved rate and therefore carries the *largest* asymptotic standard error (Fig. 3): identifiable is not the same as precisely determined. The same structure holds on the full 5,446-family fit — duplication–loss coupling 0.93, turnover the soft mode, transfer the largest standard error — with every standard error tightened by the larger sample (transfer to a $\times/1.8$ rate factor) and $\mathbf{H}_F \succ 0$ at $\lambda=1$ ($\lambda_{\min}=0.46$); we show the 256-family fit because resolving the near-zero tail requires `fp64`, impractical on all families. The bound-constrained fit corroborates this directly through its active set: 126 of the 170 box-active coordinates are duplication or loss rates driven to the lower floor (30 species have *both* δ and λ pinned — “these events $\approx$ never happen here, and the data cannot say which”), whereas transfer is mostly free with a median rate of 0.33. The non-identifiability is thus not a numerical nuisance but a biological statement about which evolutionary events the data constrain, and it is exactly what the smoothing prior of Sec. 3.5 shares strength across to resolve.

3.4 Transfer heterogeneity via per-recipient weights

[EXPERIMENT — Receiver weights]

Fit the model with and without per-recipient weights w . (a) Show that the inferred w_s identify species/lineages that disproportionately receive transfers, and relate them to known biology where possible. (b) Compare model fit (held-out likelihood and/or information criterion) with vs. without w , controlling for the added parameters. Goal: show GPUREC can express and infer across-species transfer heterogeneity — a model extension listed as future work for ALERAX — at no additional derivation cost.

3.5 Custom penalty + cross-validation on HOGENOM (the GPU-enabled study)

Table 2: Ablation of model extensions (lower held-out NLL is better). **[TODO: fill in; cross-validated, HOGENOM species-wise.]**

Model	Held-out NLL
Baseline DTL (MLE)	[TODO:]
+ tree-smoothing prior	[TODO:]
+ transfer weights	[TODO:]
+ smoothing + transfer weights	[TODO:]

[EXPERIMENT — Cross-validated tree-smoothing prior]

Fit species-wise DTL rates on HOGENOM with the tree-smoothing prior of Eq. (6). Select the penalty strength κ by K -fold cross-validation over gene families: for each κ on a grid and each fold, fit on the training families and evaluate the held-out negative log-likelihood. Report (a) the CV curve (held-out NLL vs. κ) with the selected κ^ ; (b) evidence that regularization improves held-out likelihood relative to the unpenalized MLE and reduces degenerate (boundary-saturated / non-identifiable) estimates; (c) the total compute ($\#\kappa \times K$ full fits) and wall-clock, contrasted with an estimate of the CPU time the same protocol would require. Goal: demonstrate a statistically principled, cross-validated model-selection workflow for DTL rates that is enabled by GPU throughput and would be impractical on the CPU — the headline “only-possible-because-differentiable-and-fast” result.*

Cross-validation selects a regularizing prior. On 1,055 HOGENOM families with $K=5$ -fold cross-validation over a grid of κ , the tree-smoothing prior of Eq. (6) improves held-out likelihood over the unpenalized MLE and selects $\kappa^* \approx 1$. At κ^* the fraction of boundary-saturated species-wise rates falls from ≈ 0.64 (unpenalized) to ≈ 0.05 , and — unlike the lightly-penalized $\kappa=0.03$ fit of Sec. 3.2 — the *unconstrained* optimum is already positive-definite ($\lambda_{\min} = +0.22$): the prior, not a box, supplies the missing identifiability. The entire sweep ($\#\kappa \times K$ full certified fits) runs end-to-end on a single GPU, which is what makes the protocol practical.

3.6 Ablation: which extension matters?

[EXPERIMENT — Component ablation]

Isolate the contribution of each model extension by held-out likelihood (cross-validated, same protocol as Sec. 3.5), filling Table 2. Goal: clarify how much the tree-smoothing prior and the transfer weights each contribute, and whether they are complementary.

4 Discussion

An extensible, differentiable engine. The unifying theme is that a single design choice — implementing the reconciliation likelihood as a differentiable GPU program — yields speed, second-order inference, and model flexibility together. New terms (priors, alternative recipient models, hierarchical rate structure) are written as PYTORCH expressions and require *no* hand-derived gradients or Hessians; they immediately inherit the optimizer, the HVP-based certification, and the Fisher-information tooling.

Simplicity and reproducibility. The model can be constructed and optimized in a few lines in a Jupyter notebook, with direct programmatic access to the loss, gradient, Hessian–vector product, and per-batch solver controls. This lowers the barrier to methodological experimentation and makes analyses reproducible.

From a tool to a platform. The same differentiable core that gives the results above also makes several extensions natural rather than bespoke: a Laplace approximation to the posterior over rates (Bayesian uncertainty without MCMC), hierarchical priors in which species-wise rates inherit from clade-level hyperparameters, and the use of differentiable reconciliation as the inner objective of a species-tree search. We develop these in Appendix A; collectively they reframe GPUREC as a platform for model development rather than a single fixed tool.

Limitations. (i) We address rate estimation and reconciliation given a fixed species tree and CCP inputs; species-tree search is out of scope here. (ii) Per-recipient weights and per-species rates add parameters and rely on the regularization of Sec. 3.5 to remain identifiable. (iii) GPU memory bounds the largest single batch; we mitigate this with batching and adaptive solver fidelity. **[TODO: revise after experiments.]**

5 Conclusion

We presented GPUREC, a GPU-accelerated, differentiable implementation of the UNDATEDDTL amalgamated reconciliation likelihood. Beyond an order-of-magnitude speedup for genewise rate estimation, differentiability gives certified optima, Fisher-information-based uncertainty and identifiability analysis, user-defined penalties with cross-validated model selection, and learned transfer heterogeneity — capabilities that are difficult or impossible with existing CPU, first-order tools, and that follow naturally from exposing reconciliation through a PYTORCH interface.

Availability

GPUREC is available at **[TODO: repository URL]** under **[TODO: license]**. It installs as an editable PYTORCH package (`pip install -e .`) plus a RUST preprocessing extension built with `cargo build -release`; the resulting shared library is located via the `GPUREC_PREPROCESS_PATH` environment variable. Every experiment in this paper is reproduced by a single notebook, `experiments/sanderson_cv/repro` and the following self-contained drivers, each runnable on one GPU and each writing a self-describing artifact:

- **Speed (Sec. 3.1)** — `bench_genewise_warm_rebatch.py`: per-family rates to $|Pg| < 10^{-3}$ on HOGENOM.
- **Certified optimum (Sec. 3.2) and receiver weights (Sec. 3.4)** — `converge_bounded_joint_archaea.py`: box-constrained joint (θ, α) Newton–CG with the deflated-gauge positive-definiteness certificate and the receiver-weight Fisher information.
- **Observed information (Sec. 3.3)** — `fisher_information_s53.py`: the dense $\mathbf{I}_{\text{obs}}$ spectrum, the duplication–loss confounding, and asymptotic standard errors.
- **Cross-validation (Secs. 3.5 and 3.6)** — `run_cv.py`: K -fold tree-smoothing cross-validation over a κ grid.

Datasets are the published HOGENOM and Davín et al. (2018) archaeal CCP collections (**[TODO: data DOI / download]**); each driver reads its data root from an environment variable, and the notebook prints the exact command and expected numbers for each result.

Acknowledgements

[TODO:]

A Extensions and outlook: differentiable reconciliation as a platform

The capabilities in the main text all follow from one design choice — a differentiable, GPU-resident reconciliation likelihood with exact gradients and Hessian–vector products. The same machinery makes the following extensions natural; we sketch them here as the roadmap that distinguishes GPUREC from a single fixed tool. None requires hand-derived derivatives: each is expressed as a PYTORCH term and inherits the optimizer, the matrix-free HVP, and the information-matrix tooling.

A.1 Bayesian inference by Laplace approximation

Because the Hessian at the MAP/ML estimate is available matrix-free, the posterior over rates admits a Gaussian (Laplace) approximation

$$p(\theta \mid D) \approx \mathcal{N}(\hat{\theta}, \mathbf{H}(\hat{\theta})^{-1}), \quad (7)$$

giving credible intervals and posterior correlations *without* MCMC. The required linear solves with $\mathbf{H}$ reuse the same matrix-free conjugate-gradient / Lanczos routines used for optimization and certification; one never forms or inverts the dense Hessian. This turns the point estimates that current reconciliation tools report into full uncertainty statements, and enables penalty-based (marginal-likelihood / information-criterion) model comparison between the extensions of Sec. 2.5–2.4.

A.2 Hierarchical (clade-level) rate priors

The penalty framework of Sec. 2.5 generalizes to a hierarchy: per-species rates θ_s can be drawn from clade-level hyper-parameters μ_c (themselves estimated), e.g. $\theta_s \sim \mathcal{N}(\mu_{c(s)}, \sigma^2)$ with the tree-Laplacian (Eq. (6)) as the special case of a single global level. This partial pooling shares strength among related lineages — regularizing the weakly-identified directions diagnosed in Sec. 3.3 — and is again just an additional differentiable term in $R(\theta)$, with the hyper-parameters optimized (empirical Bayes) or integrated (via the Laplace approximation of App. A.1).

A.3 Differentiable reconciliation inside species-tree search

The main text fixes the species tree S . Species-tree inference (SPECIESRAX/ALEXRAX) searches over S by SPR moves, recomputing rates at each candidate. A fast, differentiable rate-estimation inner loop is exactly the expensive primitive that search repeatedly invokes; GPUREC can serve as that inner objective, and the gradient/curvature it exposes opens the possibility of continuous relaxations of branch-level structure. We leave a full treatment to future work, but note that the engine presented here is the component such a search would be built around.

A.4 Other consequences

The same interface supports: alternative recipient models beyond the softmax of Sec. 2.4 (any differentiable parameterization); gradient-based sensitivity analysis of inferred DTL event counts to the rates; and integration with the broader differentiable-programming ecosystem (e.g. amortized or variational inference over rates). Each is a direct consequence of exposing the likelihood, its gradient, and its curvature through a standard autodiff interface.

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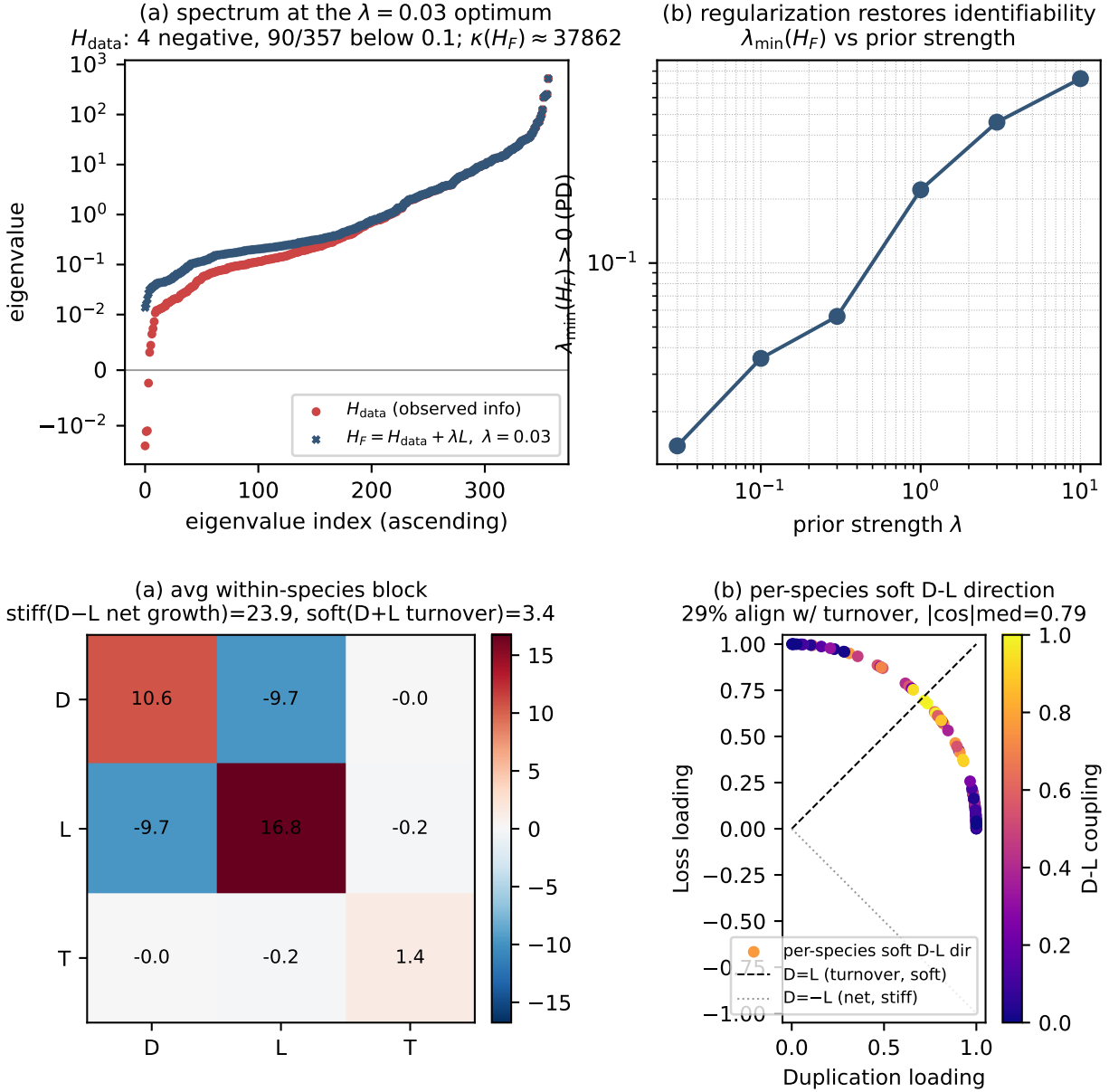


Figure 2: Observed-information analysis at an archaeal species-wise optimum (256 families, fp64-certified; $p = 3S = 357$, exact dense $\mathbf{I}_{\text{obs}}$). **Top:** (left) the data Hessian $\mathbf{I}_{\text{obs}} = \mathbf{H}_{\text{data}}$ is *indefinite* — 4 negative eigenvalues and 90/357 (25%) below 0.1 — so a quarter of directions are unconstrained by the data alone; the GBM prior λL lifts them to make $\mathbf{H}_F = \mathbf{H}_{\text{data}} + \lambda L$ positive definite. (right) $\lambda_{\min}(\mathbf{H}_F)$ climbs monotonically with prior strength ($0.014 \rightarrow 0.74$ over $\lambda = 0.03 \rightarrow 10$): regularization restores identifiability. **Bottom:** (left) the average within-species 3×3 block diagonalizes to a *stiff* net-growth axis $\delta - \lambda$ (eig 23.9) and a *soft* turnover axis $\delta + \lambda$ (eig 3.4); transfer is decoupled. (right) per species, the soft duplication–loss direction (coloured by D–L coupling) clusters on the turnover diagonal $\delta = \lambda$ — strongly-coupled species sit exactly on it.

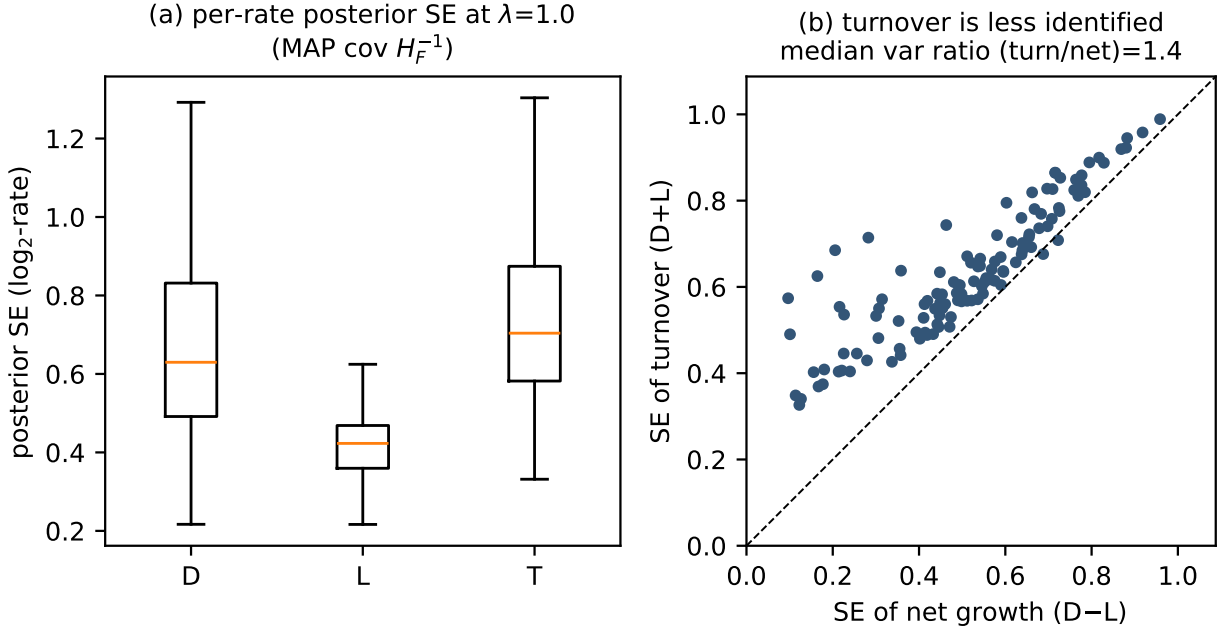


Figure 3: Asymptotic uncertainty from $\mathbf{I}_F^{-1}$ at the cross-validated optimum ($\lambda = 1$, where $\mathbf{H}_F \succ 0$). (left) per-rate posterior standard errors (log₂-rate): median $\delta = 0.63$, $\lambda = 0.42$, $\tau = 0.70$ (95% CI factors $\times / 2.35, 1.78, 2.60$). Transfer, though decoupled, carries the *largest* SE — it is identifiable but only gently determined. (right) the turnover combination $\delta + \lambda$ has $1.4\times$ the posterior variance of net growth $\delta - \lambda$ for *every* species: turnover is the least-identified direction even after regularization.

[TODO: Fig. 2: held-out NLL vs. penalty strength κ (CV curve), κ^* marked.]

Figure 4: Cross-validated selection of the tree-smoothing penalty for species-wise rates on HOGENOM. [TODO: caption.]